FUZZY LAB - 1

Matrix and Vector Operations

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```
%% 1. Matrix and Vector Construction

m = 3;
n = 4;

A = ones(m,n)
```

```
A = 3×4
     1     1     1     1
     1     1     1     1
     1     1     1     1
```

```
v = ones(n,1)
```

```
v = 4×1
     1
     1
     1
     1
```

```
%% 2. Random Matrices

A_uniform = rand(m,n)
```

```
A_uniform = 3×4
    0.3371    0.3112    0.6020    0.6892
    0.1622    0.5285    0.2630    0.7482
    0.7943    0.1656    0.6541    0.4505
```

```
A_normal = randn(m,n)
```

```
A_normal = 3×4
   -2.1384   -1.0722    1.4367   -1.2078
   -0.8396    0.9610   -1.9609    2.9080
    1.3546    0.1240   -0.1977    0.8252
```

```
A_int = randi([1 100],m,n)
```

```
A_int = 3×4
    87    26    92    15
     9    81    19    14
    40    44    27    87
```

```
%% 3. Sum of Rows, Columns and Total

A = randi([1 10],m,n)
```

```
A = 3×4
     6     9     6     3
     6     7     5     2
     2     4     1     2
```

```
row_sum = sum(A,2)
```

```
row_sum = 3×1
    24
    20
     9
```

```
col_sum = sum(A,1)
```

```
col_sum = 1×4
    14    20    12     7
```

```
total_sum = sum(A(:))
```

```
total_sum = 
53
```

```
%% 4. Mean and Median

v = randi([1 20],10,1)
```

```
v = 10×1
     5
     9
     1
    19
    19
    10
    10
     7
    19
     8
```

```
mean(v)
```

```
ans = 
10.7000
```

```
median(v)
```

```
ans = 
9.5000
```

```
%% 4. Mean and Median

v = randi([1 20],10,1)
```

```
v = 10×1
     3
    16
     8
     5
     9
     2
     3
    19
    20
    12
```

```
mean(v)
```

```
ans =
9.7000
```

```
median(v)
```

```
ans =
8.5000
```

```
%% 5. Size of Matrix and Vector

size(A)
```

```
ans = 1×2
     3     4
```

```
size(v)
```

```
ans = 1×2
    10     1
```

```
%% 6. Logical Indexing

A = randi([1 20],3,3)
```

```
A = 3×3
     2    17     4
     5     1    13
     8     1    15
```

```
A(A>10)
```

```
ans = 3×1
    17
```

```
    13
    15
```

```
A(A<10)
```

```
ans = 6×1
     2
     5
     8
     1
     1
     4
```

```
%% 7. Inverse of Matrix

B = randi([1 5],3,3)
```

```
B = 3×3
     4     2     4
     3     4     1
     3     1     2
```

```
if det(B) ~= 0
    inv(B)
else
    disp('Matrix is singular')
end
```

```
ans = 3×3
   -0.5000         0    1.0000
    0.2143    0.2857   -0.5714
    0.6429   -0.1429   -0.7143
```

```
%% 8. Special Matrices

diag([1 2 3 4])
```

```
ans = 4×4
     1     0     0     0
     0     2     0     0
     0     0     3     0
     0     0     0     4
```

```
eye(4)
```

```
ans = 4×4
     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1
```

```
zeros(3,4)
```

```
ans = 3×4
     0     0     0     0
     0     0     0     0
     0     0     0     0
```

```
%% 9. Matrix Multiplication

A1 = randi([1 5],3,2)
```

```
A1 = 3×2
     4     5
     4     4
     1     3
```

```
B1 = randi([1 5],2,4)
```

```
B1 = 2×4
     3     2     3     4
     3     3     5     4
```

```
A1 * B1
```

```
ans = 3×4
    27    23    37    36
    24    20    32    32
    12    11    18    16
```

```
%% 10. Sorting

A = randi([1 20],3,3)
```

```
A = 3×3
     8     8    12
    17    19    13
    11    18    12
```

```
sort(A)
```

```
ans = 3×3
     8     8    12
    11    18    12
    17    19    13
```

```
sort(A(:))
```

```
ans = 9×1
     8
     8
    11
    12
    12
    13
    17
```